

Open Economy Project

Title: Exchange Rate Volatility and Its Impact on Capital Flows: A Comparative Analysis of India and Japan

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Introduction

Research question:

The objective of the research is to analyze the relationship between exchange rate volatility, interest rates and capital flows between India and Japan. Through this research, we have attempted to relate the theoretical concepts in the exchange rate determination model to the real world, using these countries as an example.

Hypothesis:

Through this research, we are attempting to study if the models of exchange rate determination and relationships between variables (inflation, interest rate, capital movements and exchange rates) we have studied so far, can be observed in the real world. To test this hypothesis, we have selected India and Japan as the two countries.

We have conducted regression and correlation analysis of FDI inflows, outflows, inflation rates, exchange rates and investment rates as well as graphical representation of the given data.

Significance:

This project explores how exchange rate volatility affects capital flows between India and Japan, shedding light on key economic dynamics. By examining this relationship, it provides insights valuable for policymakers aiming to stabilize currency and attract investment, supporting India's growth. It also applies economic theories, such as Uncovered Interest Parity (UIP), to real-world data, enriching both academic understanding and practical investment strategies by highlighting currency risks. Ultimately, the project's findings are relevant for shaping economic policy, enhancing investment appeal, and supporting long-term stability.

Data description and Methodology:

Data is collected from different websites and organized in an excel sheet. The links to these sites are provided in the References section. The data is taken yearly over the past 11 years. This is because monthly data was not available for both the countries. Net inflows for foreign direct investment and portfolio investment are taken for both the countries to comprehend overall effect.

Interest rate differential is calculated by deducting the repo rate for India from that of Japan.

Interest rate differential = repo rate for Japan - repo rate for India

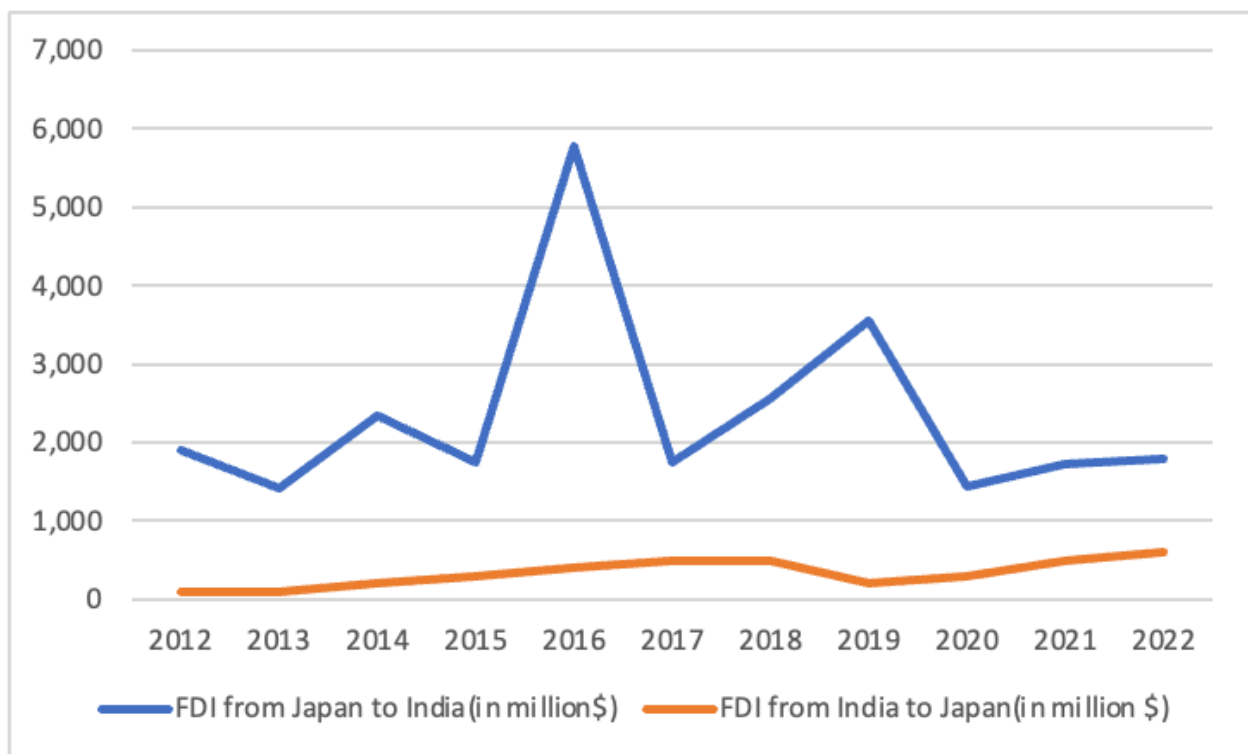
Regression and correlation analysis is carried out using Microsoft Excel.

Results and Analysis:

Foreign direct investments:

According to the OECD (Organisation for Economic Co-operation and Development), Foreign Direct Investment (FDI) is defined as "a category of cross-border investment associated with a resident in one economy having control or a significant degree of influence on the management of an enterprise that is resident in another economy." Typically, this requires the investor to hold at least 10% of the voting power in the foreign enterprise, marking it as a long-term investment aimed at establishing a lasting interest.

The OECD emphasizes FDI as an important driver of economic integration, bringing capital, technology, and expertise across borders.



Foreign direct investment inflows between India and Japan between the years 2012-2022

The foreign direct investment inflows between India and Japan have increased over the years. Current FDI inflows to India during these years is approximately \$21 billion while cumulative FDI inflows in Japan is approximately \$2 billion (as of 2023)

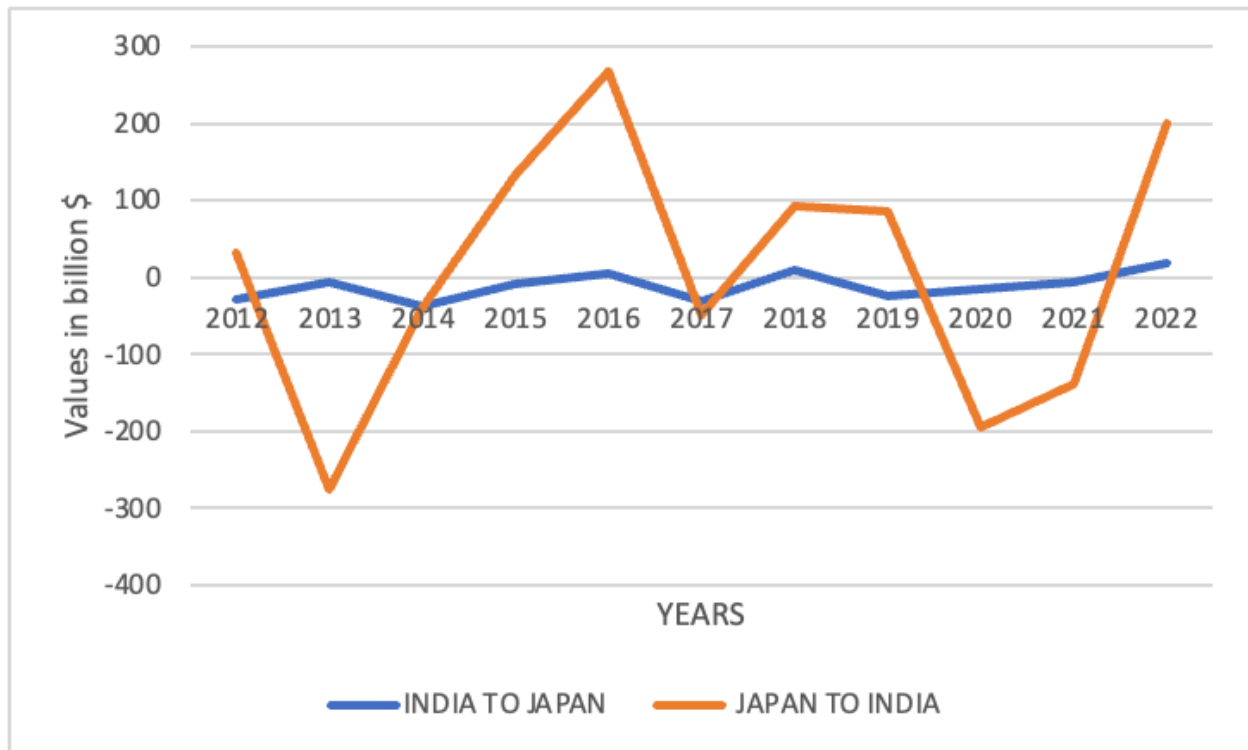
In the year 2016, India experienced the all-time high FDI from Japan (\$5.7 billion). The peak in Japan's capital flow to India can be attributed to various reasons such as India's big and growing market and resources, diversification from China, etc. India has also invested heavily in Japan over the years as a part of technological collaboration, strategic partnership, etc.

	INDIA (values in million \$)	JAPAN (values in million \$)
MEAN	2290.42	336.36
MEDIAN	1854.5	300
STANDARD DEVIATION	1198.8	174.77
MAXIMUM	5781	100
MINIMUM	1295	600

Portfolio investments net inflows:

Portfolio investments involve purchasing financial assets, like stocks, bonds, or other securities, in a foreign country without seeking direct control or influence over the business. Unlike Foreign Direct Investment (FDI), which is focused on gaining a lasting interest and influence, portfolio investments are more passive. They aim to generate returns from dividends, interest, or capital appreciation rather than participate in day-to-day management.

Examples include buying shares of a foreign company listed on a stock exchange or purchasing government bonds of another country. Portfolio investments are typically more liquid and can be sold more easily than FDI assets, making them a popular choice for investors seeking diversification without committing to long-term business involvement abroad.



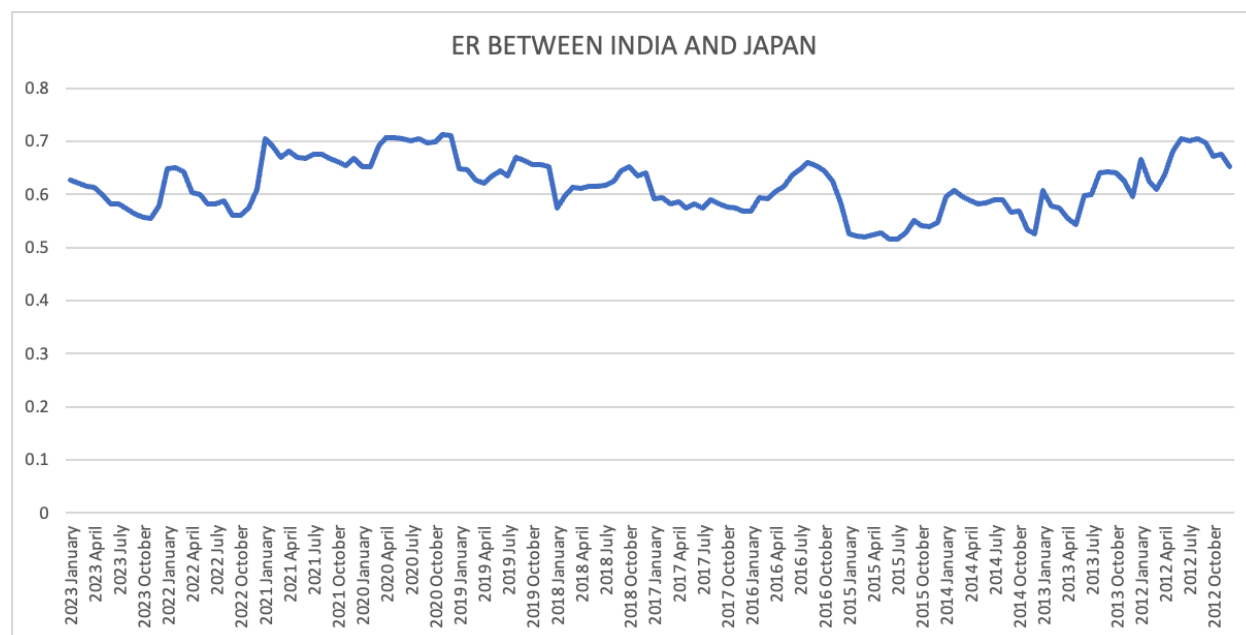
From the above graph we can interpret that portfolio investments from Japan to India has been between -300 to +300 billion US dollars while net inflows in Japan have remained negative for most of the years. Japanese investors have invested heavily in Indian market due to India's reforms and ease of doing business.

	INDIA	JAPAN
MEAN	10.13	-13.08
MEDIAN	32.2	-9
STANDARD DEVIATION	167.8	17.67
MAXIMUM	19	268
MINIMUM	-38	-275

EXCHANGE RATE:

An exchange rate is the value at which one currency can be exchanged for another. It tells you how much of one currency (like the US dollar) is needed to buy a unit of another currency (like the euro). Exchange rates fluctuate based on supply and demand

in the foreign exchange market and can be influenced by factors like interest rates, inflation, economic stability, and geopolitical events.



The exchange rate between India and Japan between the years 2012-2023

The Indian Rupee (INR) is managed by the Reserve Bank of India, while the Japanese Yen (JPY) is regulated by the Bank of Japan. As of 26 October, 2024 exchange rate is 1 JPY = 0.55 INR.

From 2012 to 2023, a few key trends and patterns stand out. Each year shows seasonal fluctuations, with certain months consistently seeing higher or lower values. Between 2020 and 2023, there is a clear downward trend, with 2023 displaying noticeably lower values compared to previous years. Over the longer term, values have declined slightly, starting from an average of around 0.68 in 2012 and gradually dropping to between 0.55 and 0.62 by 2023.

A peak was observed around 2020, but after that, values consistently declined year after year. For individual years, the highest values tend to occur in the colder months like November and December, while summer months usually see lower values. Notably, in recent years (2021-2023), values in January and February have been higher compared to other months in the same year. January often begins strong, with a gradual drop by mid-year, while December values vary—sometimes rising, as in 2022 and 2020, and sometimes falling below January levels.

India operates a managed floating exchange rate system, where the Reserve Bank of India may intervene in the currency market to control inflation and prevent excessive volatility. This intervention helps stabilize the Rupee in response to economic shifts.

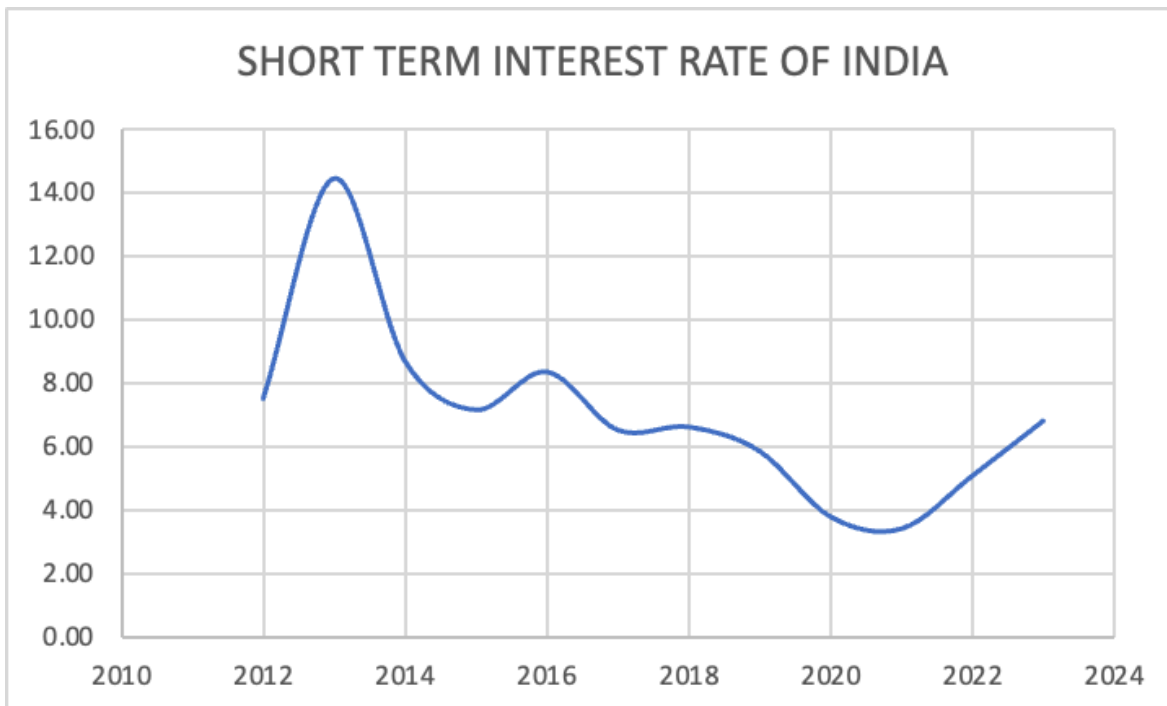
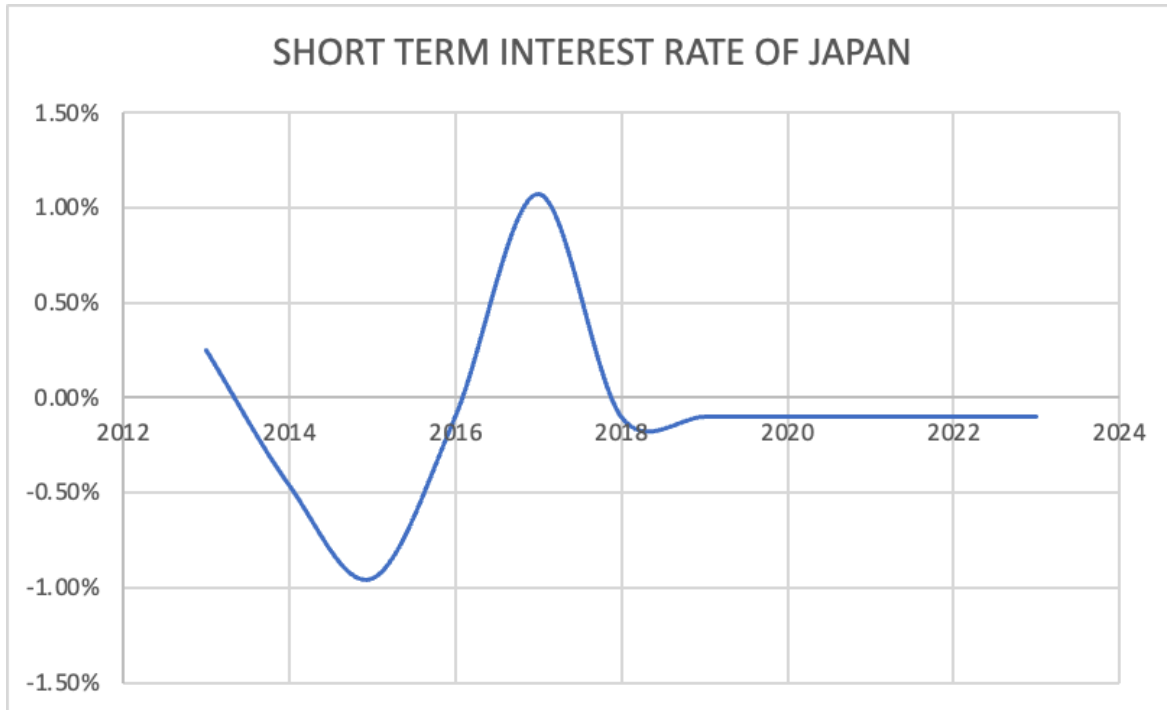
From 2012 to 2024, the INR has gradually depreciated against the JPY, with the exchange rate moving from 1 JPY = 0.525 INR in 2012 to 0.69 INR in 2024. This trend reflects a strengthening of the Yen relative to the Rupee, influenced by factors such as inflation rates, economic policies, and trade balances between the two countries.

This consistent rise in the exchange rate highlights the impact of economic conditions and monetary policies on currency values over time.

MEAN	0.614723
MEDIAN	0.61466
STANDARD DEVIATION	0.051019
MAXIMUM	0.711916
MINIMUM	0.515493

Repo Rates

The repo rate is the interest rate at which a central bank, like the Reserve Bank of India (RBI), lends money to commercial banks for short-term needs. When the RBI raises the repo rate, borrowing becomes more expensive, which helps control inflation by reducing spending. Lowering the repo rate makes borrowing cheaper, encouraging spending and investment in the economy. It's a key tool for managing inflation and economic growth.



The repo rates for India and Japan have been analyzed on a bimonthly basis to facilitate a fair comparison, given that data for both countries was available for alternate months. This method allows for a clearer visualization of trends and economic policies.

In India, the repo rates began at a remarkably high 14.45% in 2013, a measure taken to combat inflationary pressures. This high rate persisted until 2014, after which there was a gradual decline, indicating a shift towards promoting economic growth. The years leading up to the Covid-19 pandemic saw rates stabilize, with a peak of 6.6% in 2018. However, the onset of the pandemic in 2020 prompted a sharp decrease to 3.76% as the Reserve Bank of India sought to stimulate the economy during a time of crisis. Following a period of recovery, the repo rate increased to 6.80% in 2023, reflecting a response to rising inflation and a need to balance growth with price stability.

Japan's repo rates illustrate a different monetary policy trajectory, starting at 0.25% in 2013 and quickly moving into negative territory by 2014. This policy aimed to encourage borrowing and spending amid prolonged low growth and deflationary pressures. The negative rates remained consistent, reflecting Japan's ongoing economic challenges and the difficulties in achieving stable inflation and growth. The brief rise to 1.07% in 2017 was a temporary shift that quickly reverted, indicating the fragility of Japan's economic recovery.

Comparative Overview:

Comparing the two countries, India has experienced significant fluctuations in its repo rates, adapting its monetary policy to manage inflation while responding to economic crises. In contrast, Japan has maintained a low and negative interest rate environment for nearly a decade, reflecting persistent economic stagnation and an inability to generate inflation.

Overall, the repo rates for both India and Japan highlight the complexities of their respective economic landscapes and the diverse strategies employed by their central banks. India's approach showcases a more dynamic response to economic conditions, while Japan's prolonged negative rates reveal the challenges of stimulating growth in a struggling economy.

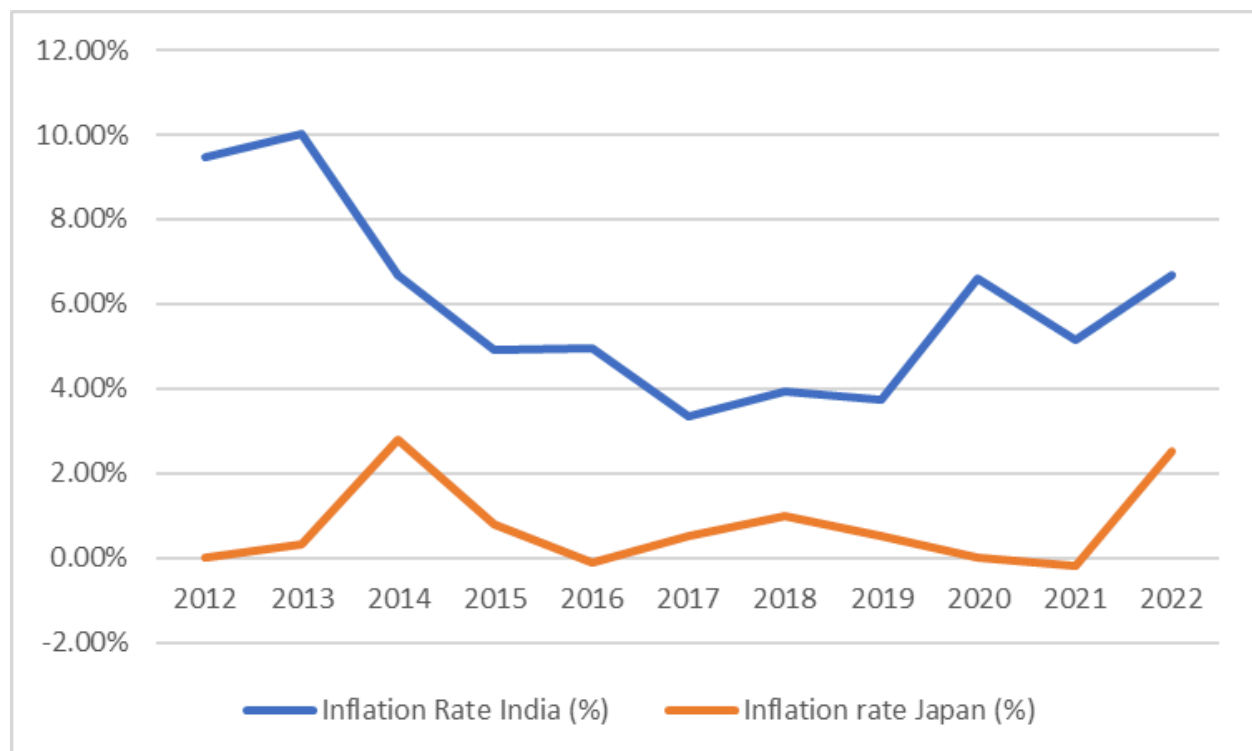
	INDIA	JAPAN
MEAN	0.0325	-0.0005
MEDIAN	0.0357	-0.001
STANDARD DEVIATION	0.03	0.0046
MAXIMUM	0.142	-0.001
MINIMUM	0.03	0.25

Consumer Price Index

The Consumer Price Index (CPI) is an economic indicator that measures the average change over time in the prices paid by consumers for a fixed basket of goods and

services. It includes a variety of categories, such as food, housing, clothing, transportation, medical care, and entertainment. The CPI is used to assess inflation and reflects how price changes affect the purchasing power of consumers.

Calculating the CPI involves comparing the current cost of the basket to the cost in a base year. The index is crucial for economic analysis, influencing monetary policy decisions, adjusting wages, and indexing social security payments. By tracking changes in the CPI, policymakers and economists can better understand economic conditions and consumer behavior.



The consumer price index for India and Japan plotted yearly from 2012-2023

India has experienced significant fluctuations in the inflation rates in the past couple of years on the other hand inflation rate in Japan has remained steady and very low. Recent increase in Infaltion rates of both countries can be attributed to covid-19 pandemic and global economy slowdown.

	INDIA	JAPAN
MEAN	5.95%	1%

MEDIAN	5.13%	1%
STANDARD DEVIATION	0.022	0.01
MAXIMUM	10.02%	-0.20%
MINIMUM	3.33%	2.80%

Correlation interpretation

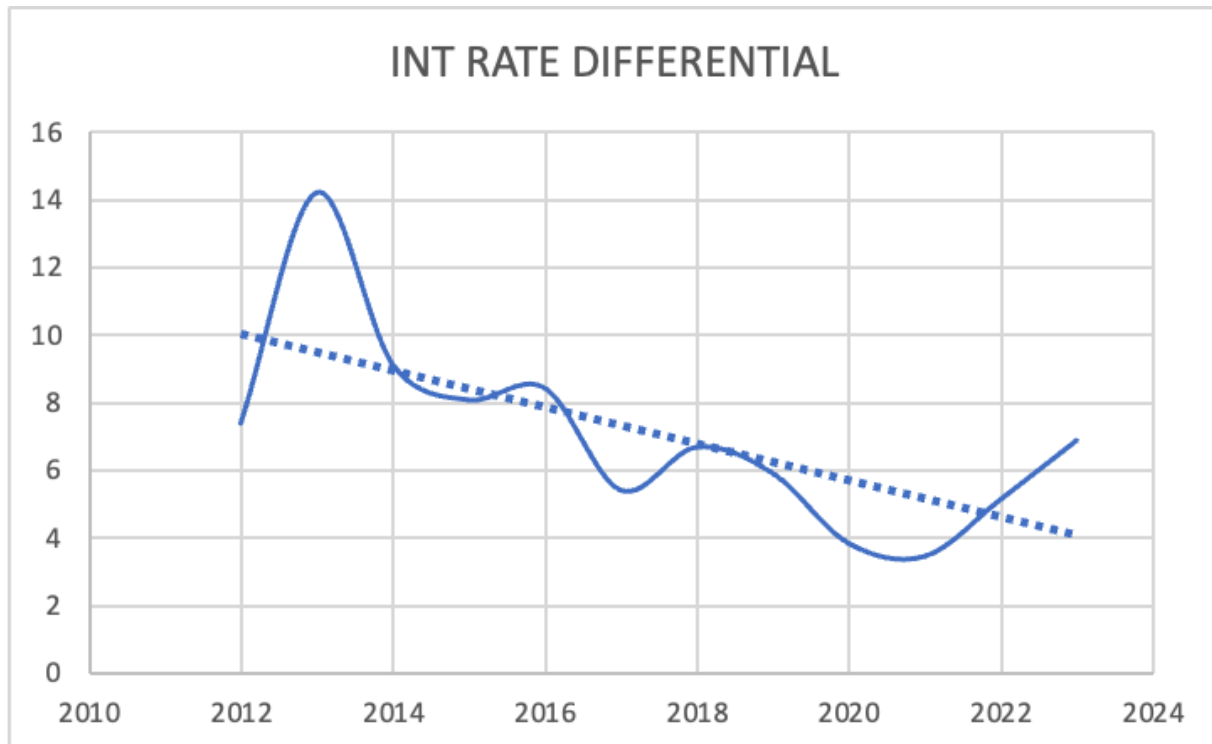
Correlation between exchange rate and inflation rates of both countries

COUNTRY	CORRELATION
INDIA	-0.502
JAPAN	0.07

The correlation between exchange rates and inflation varies significantly between countries. Generally, when a country experiences higher inflation than another, its currency tends to depreciate. This is due to the purchasing power parity (PPP) theory, which suggests that currencies adjust to maintain price level parity across countries

Here the correlation between India's inflation rate and exchange rate is -0.502 which indicates a moderate negative relationship between the two variables. On the other hand, correlation between inflation rate of Japan and exchange rate is 0.07 which indicates no significant correlation between the two.

Relating the interest rate differential to exchange rates



The interest rate differential here is the difference in repo rates in INDIA and JAPAN, such that interest rate differential = i^* (India) - i (Japan).

Analyzing the interest rate differential between India and Japan from 2012 to 2023, we can see how the differential provides potential for both uncovered and covered interest arbitrage, impacting currency value and investment flows. With India's higher rates, the differential suggests that investors could profit by borrowing in Japan (where rates are near zero) and investing in India. For example, in 2013, the differential peaked at 14.2%, creating a strong incentive for arbitrage that would cause the INR to depreciate relative to the JPY, as predicted by the uncovered interest arbitrage equation:

$$i = i^* + (E^e - E)/E$$

In this equation, (i) represents India's domestic interest rate, (i^*) is Japan's lower foreign rate, (E) is the spot exchange rate (INR/JPY), and (E^e) represents the expected exchange rate. When the differential is high, investors expect INR depreciation in line with the differential, allowing them to profit despite the lower Japanese interest rate.

In years with lower differentials, such as 2021 (3.48%) and 2020 (3.86%), uncovered interest arbitrage may become less attractive since the potential gain from the rate difference is smaller, and the INR might not be expected to depreciate significantly. However, in moderately high-differential years like 2016 (8.44%) and 2018 (6.7%), investors might prefer covered

interest arbitrage to hedge exchange rate risk by fixing the rate through a future contract. In this scenario, the covered interest arbitrage equation applies:

$$i = i^* + (F - E)/E$$

where (F) is the future exchange rate (hedged). By locking in an exchange rate, investors gain from India's higher interest rate without risking fluctuations in INR/JPY.

The theory suggests that if the differential favors the foreign market, the home currency (INR) should depreciate approximately equal to the differential over time. In years with larger differentials, we can expect more depreciation pressure on the INR as investors seek higher returns in India. Conversely, smaller differentials might reduce this pressure, contributing to more stable exchange rates. Thus, the interest rate differential data from 2012 to 2023 not only reflects investor strategies but also aligns with theoretical expectations, driving INR depreciation relative to the JPY in high-differential years and moderating flows when the differential is narrower.

The correlation between the exchange rate and interest rate differential is 0.44 which is a moderate correlation. The results of the regression are as follows:

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.681507262							
R Square	0.464452148							
Adjusted R Square	0.410897363							
Standard Error	0.041510543							
Observations	12							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	0.014943748	0.014943748	8.672467761	0.014661918			
Residual	10	0.017231252	0.001723125					
Total	11	0.032175						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.51607945	0.033276127	15.5089997	2.53607E-08	0.441935618	0.590223283	0.441935618	0.590223283
X Variable 1	0.012947558	0.004396596	2.944905391	0.014661918	0.003151333	0.022743784	0.003151333	0.022743784

Here the exchange rate is the dependent variable and interest rate differential is the independent variable. The R square value is 0.4644 which shows that 46.44% variation in exchange rate is explained by interest rate differential. This shows that exchange rates are influenced by interest rate differentials to a moderate extent.

In our example, the Indian rupee is an expensive currency in relation to the Japanese yen. Also the interest rates are higher in India as compared to Japan . So as the interest rate differential rises it leads to an appreciation of the Indian currency which can be interpreted through the model.

Relating capital flows to exchange rates

SUMMARY OUTPUT									
<i>Regression Statistics</i>									
Multiple R	0.745831								
R Square	0.556264								
Adjusted R	0.44533								
Standard Error	0.037051								
Observations	11								
<i>ANOVA</i>									
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>				
Regression	2	0.013768	0.006884	5.014371404	0.038770212				
Residual	8	0.010982	0.001373						
Total	10	0.02475							
<i>Coefficients</i>									
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>	
Intercept	0.535639	0.032347	16.55895	1.78642E-07	0.461045759	0.610232	0.461046	0.610232	
INDIA	-3E-06	9.11E-06	-0.32798	0.751348473	-2.39948E-05	1.8E-05	-2.4E-05	1.8E-05	
JAPAN	0.000212	6.71E-05	3.162923	0.013336088	5.75318E-05	0.000367	5.75E-05	0.000367	

Exchange rate is the dependent variable and net FDI inflow is the independent variable.

Here the R square value is 0.55 and adjusted r square is 0.44 which shows that variations in exchange rate is explained by the Foreign direct investment by both countries to some extent. As the foreign direct investment by Japan in India increases the Indian rupee appreciates and yen depreciates. There is a positive correlation of 0.46 between the exchange rate and FDI which also proves the point.

This regression analysis evaluates the relationship between exchange rates in Japan and India and their effect on Foreign Direct Investment (FDI) flows. By examining the exchange rates as predictors of FDI, the model suggests a moderate connection between currency value fluctuations and investment activity.

The results indicate that around 55.6% of the variation in FDI can be explained by the exchange rates of the Japanese yen and Indian rupee. This level of explanatory power implies that exchange rates influence FDI decisions to a notable extent, though other external factors likely contribute significantly to investment patterns as well.

Within this model, the Japanese yen's exchange rate shows a statistically significant relationship with FDI. The positive coefficient associated with Japan's exchange rate implies that an appreciation in the yen may encourage higher FDI inflows, potentially because a stronger yen might signal economic stability or increased purchasing power, making Japan an attractive investment destination.

In contrast, the Indian rupee's exchange rate does not appear to have a significant influence on FDI. This lack of statistical significance suggests that currency fluctuations for India may not strongly impact FDI levels, possibly due to other overriding factors like economic policy, market size, or business environment which play a larger role in shaping FDI patterns.

Overall, the analysis indicates that Japan's currency fluctuations are more influential on FDI in this model than India's, with the yen's value changes potentially guiding investment decisions more directly. This finding highlights the importance of considering country-specific factors in evaluating the impact of exchange rates on international investment.

<i>Regression Statistics</i>									
Multiple R	0.479833								
R Square	0.23024								
Adjusted R Square	0.037799								
Standard Error	0.0488								
Observations	11								
<i>ANOVA</i>									
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>				
Regression	2	0.005698	0.002849	1.196422	0.351093				
Residual	8	0.019052	0.002381						
Total	10	0.02475							
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>	
Intercept	0.616054	0.018353	33.56632	6.78E-10	0.573731	0.658376	0.573731	0.658376	498
japan	0.001381	0.00092	1.500658	0.171837	-0.00074	0.003503	-0.00074	0.003503	017
INDIA	-2.3E-05	9.96E-05	-0.23311	0.821531	-0.00025	0.000207	-0.00025	0.000207	541

R squared value is 0.23 which means that only 23% variations in exchange rate is explained by portfolio investments. Moreover the correlation is 0.24 which means that portfolio investments are not a good indicator of exchange rate volatility.

This regression analysis explores the relationship between exchange rates of Japan and India and their potential impact on portfolio investments between the two countries. Based on the model's findings, the connection between exchange rates and portfolio investment appears relatively weak, suggesting that other factors likely play a more significant role in influencing investment flows.

The R-Square value of 0.230 indicates that only about 23% of the variation in portfolio investment can be explained by the exchange rates of the Japanese yen and Indian rupee combined. This relatively low explanatory power implies that currency fluctuations may not strongly drive portfolio investment decisions in this case. The adjusted R-Square, which accounts for the number of predictors, is even lower at 0.037, further reinforcing the modest influence of exchange rates on portfolio investment.

The ANOVA significance test, with a p-value of 0.351, shows that the overall model is statistically insignificant at conventional levels. This suggests that the exchange rates of Japan and India, as captured by this model, are not reliable predictors of portfolio investment between these countries.

Looking at the individual coefficients, the exchange rate of Japan shows a positive coefficient (0.00138), indicating that an appreciation in the yen could have a slight positive effect on portfolio investment. However, this effect is not statistically significant (p-value = 0.172), meaning we cannot conclude with confidence that yen fluctuations meaningfully impact portfolio investment.

Similarly, the Indian rupee's coefficient is negative (-0.000023) but also statistically insignificant (p-value = 0.822). This result suggests that changes in the Indian rupee's exchange rate do not have a meaningful impact on portfolio investment within this model.

In summary, this analysis implies that exchange rates for Japan and India may have minimal influence on portfolio investment flows between the two countries. Other factors, such as economic growth, interest rates, or political stability, are likely more significant drivers of investment activity in this context.

Relating inflation to exchange rate

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.50912							
R Square	0.259204							
Adjusted R Square	0.074004							
Standard Error	0.047873							
Observations	11							
<i>ANOVA</i>								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	2	0.006415	0.003208	1.399594	0.301159			
Residual	8	0.018335	0.002292					
Total	10	0.02475						
<i>Coefficients</i>								
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.664401	0.044461	14.94353	3.97E-07	0.561875	0.766928	0.561875	0.766928
X Variable 1	-1.12948	0.682916	-1.65391	0.136745	-2.70429	0.445328	-2.70429	0.445328
X Variable 2	0.38478	1.484107	0.259267	0.801976	-3.03758	3.807136	-3.03758	3.807136

Here the R square is 0.25 which shows that 25% variations in exchange rate is explained by the inflation rate in both countries. As inflation rises in a country its currency depreciates which can be seen in the case of Japan but not India which can be due to policy interventions and other measures.

This regression analysis examines the potential relationship between inflation rates and exchange rates, seeking to understand how inflationary changes in two variables (denoted as X Variable 1 and X Variable 2) might influence exchange rate behavior. However, the model indicates a weak relationship, suggesting that inflation rates in this context may not be substantial drivers of exchange rate fluctuations.

The R-Square value of 0.259 suggests that only about 25.9% of the variation in exchange rates can be explained by the inflation-related variables included. This relatively low explanatory power implies that factors beyond inflation might play a more prominent role in determining exchange rates. Furthermore, the adjusted R-Square of 0.074, which accounts for the number of predictors, reflects an even smaller influence, suggesting that the impact of inflation on exchange rates is minimal within this model.

The overall model is statistically insignificant, as indicated by an F-statistic p-value of 0.301. This means that, collectively, the predictors do not have a reliable effect on exchange rates at a conventional level of significance.

Examining the individual coefficients provides further insight. The coefficient for X Variable 1 is -1.129, which indicates a weak negative relationship with exchange rates. However, this coefficient is statistically insignificant (p-value = 0.137), meaning we cannot confidently conclude that changes in X Variable 1 meaningfully influence exchange rates. Similarly, X Variable 2 has a positive coefficient of 0.385, suggesting a minor positive relationship, yet it too is statistically insignificant (p-value = 0.802).

In summary, the analysis suggests that inflation, as represented by these variables, has limited explanatory power over exchange rate changes in this model. Other factors, such as interest rates, economic policies, or geopolitical stability, may be more influential in determining exchange rates, with inflation appearing to have only a minimal impact here.

Conclusion

In summary, this analysis highlights the limited impact of economic factors like interest rate differentials, portfolio investment, FDI, and inflation rates on exchange rate movements between Japan and India. While these variables are often expected to significantly influence currency fluctuations, the managed nature of both the Japanese yen and Indian rupee may temper their responsiveness to such drivers, resulting in only a modest relationship.

The findings indicate that exchange rates between Japan and India are likely influenced by a broader array of complex factors beyond those traditionally considered in theoretical models, such as regulatory policies, geopolitical dynamics, and global market sentiment. This complexity may explain why the relationships do not strictly adhere to established economic theories.

As Japan and India continue to grow as prominent players in the global economy, it will be essential to consider additional influences beyond interest rates, inflation, and investment flows when analyzing their currency dynamics. This perspective underscores the unique nature of emerging and advanced economy interactions, where structural and policy differences can shape exchange rate behavior in ways that traditional models may not fully capture.

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Appendix

I. Short Term Interest Rate For India and Japan

Year	India	Japan
2023	6.80	-0.10%
2022	5.06	-0.10%
2021	3.38	-0.10%
2020	3.76	-0.10%
2019	5.83	-0.10%
2018	6.6	-0.10%
2017	6.5	1.07%
2016	8.34	-0.10%
2015	7.14	-0.95%
2014	8.68	-0.46%
2013	14.45	0.25%
2012	7.5	0.1%

II. Inflation Rate

Year	India	Japan
2012	0	5.65
2013	0.3	6.7
2014	2.8	5.13
2015	0.8	6.62
2016	-0.1	3.73
2017	0.5	3.94
2018	1	3.33
2019	0.5	4.95
2020	0	4.91
2021	-0.2	6.67
2022	2.5	10.02
2023	3.3	9.48

III. Portfolio Investment Of India and Japan (in US\$)

Year	India	Japan
2023	-31 B	199 B
2022	19 B	(138) B
2021	-6 B	(196) B
2020	-15 B	35.9 B
2019	-25 B	86.6 B
2018	10 B	92.2 B
2017	-31 B	(49.2) B
2016	5 B	268 B
2015	-9 B	132 B
2014	-38 B	(40.3) B
2013	-7 B	(275) B
2012	-29 B	32.2 B

IV. Interest Rate Differential of India and Japan

Year	Interest Rate Differential (India - Japan)
2023	6.90%
2022	5.16%
2021	3.48
2020	3.86
2019	5.93

2018	6.7
2017	5.43%
2016	8.44%
2015	8.09
2014	9.14%
2013	14.20%
2012	7.400%

V. Foreign Direct Investment (in millions in US\$)

Year	India	Japan
2012	100	\$1,909
2013	200	\$1,420
2014	300	\$2,335
2015	400	\$1,739
2016	500	\$5,781
2017	500	\$1,745
2018	200	\$2,558
2019	300	\$3,561
2020	500	\$1,445
2021	600	\$1930
2022	600	\$1490
2023		\$1800

VI. Bilateral Exchange Rate (Yen/INR)

Year Months	Value
2023 January	0.62791 9
2023 February	0.62110 3
2023 March	0.61525 7
2023 April	0.61425 4
2023 May	0.59995 4
2023 June	0.582
2023 July	0.58189 7
2023 August	0.57246 2
2023 September	0.56211 4
2023 October	0.55675
2023 November	0.55543 7
2023 December	0.57851
2022 January	0.6489
2022 February	0.6502

Year Months	Value
2022 March	0.64341 7
2022 April	0.60441 5
2022 May	0.59982 5
2022 June	0.5827
2022 July	0.58269 7
2022 August	0.58814 7
2022 September	0.5612
2022 October	0.56015
2022 November	0.57392 6
2022 December	0.61050 8
2021 January	0.70499 4
2021 February	0.69084
2021 March	0.66987 9
2021 April	0.68217 1
2021 May	0.67091 8

Year Months	Value
2021 June	0.66803 5
2021 July	0.67562 6
2021 August	0.67521 8
2021 September	0.66798 5
2021 October	0.66257 4
2021 November	0.6536
2021 December	0.6684
2020 January	0.65248 5
2020 February	0.6516
2020 March	0.69208 7
2020 April	0.70795 5
2020 May	0.70641 9
2020 June	0.7043
2020 July	0.70175 8
2020 August	0.70441 8

Year Months	Value
2020 September	0.69642 5
2020 October	0.69900 9
2020 November	0.71191 6
2020 December	0.71049 7
2019 January	0.64817
2019 February	0.64595 8
2019 March	0.62678 3
2019 April	0.62160 6
2019 May	0.63408 5
2019 June	0.6437
2019 July	0.63537 4
2019 August	0.66954 6
2019 September	0.66433 2
2019 October	0.65692 5
2019 November	0.65676

Year Months	Value
2019 December	0.65213 6
2018 January	0.5753
2018 February	0.59644 7
2018 March	0.61368 2
2018 April	0.61066 1
2018 May	0.61533 3
2018 June	0.61591 9
2018 July	0.61681 5
2018 August	0.62571 1
2018 September	0.64402 3
2018 October	0.65269 5
2018 November	0.63540 1
2018 December	0.6398
2017 January	0.59125 1
2017 February	0.5945

Year Months	Value
2017 March	0.583
2017 April	0.58618 9
2017 May	0.5742
2017 June	0.58146 7
2017 July	0.57372 5
2017 August	0.5893
2017 September	0.58178 1
2017 October	0.57656 5
2017 November	0.57470 2
2017 December	0.56886 8
2016 January	0.56773 4
2016 February	0.59460 6
2016 March	0.59252
2016 April	0.6053
2016 May	0.61506 6
2016 June	0.63704 3

Year Months	Value
2016 July	0.6468
2016 August	0.66090 1
2016 September	0.65404 1
2016 October	0.64400 4
2016 November	0.62494 8
2016 December	0.58454 5
2015 January	0.52490 4
2015 February	0.52267 1
2015 March	0.51920 9
2015 April	0.52410 9
2015 May	0.52760 2
2015 June	0.51549 3
2015 July	0.51596 5
2015 August	0.52788 1
2015 September	0.55136 4

Year Months	Value
2015 October	0.54125
2015 November	0.53957 3
2015 December	0.54697 7
2014 January	0.59660 8
2014 February	0.60846 4
2014 March	0.5968
2014 April	0.5886
2014 May	0.5822
2014 June	0.58475 6
2014 July	0.59069
2014 August	0.59087 2
2014 September	0.56701 3
2014 October	0.56812 4
2014 November	0.5336
2014 December	0.52525 6
2013 January	0.6085
2013 February	0.5783

Year Months	Value
2013 March	0.574
2013 April	0.5547
2013 May	0.5437
2013 June	0.5971
2013 July	0.5994
2013 August	0.6416
2013 September	0.6434
2013 October	0.64
2013 November	0.6251
2013 December	0.5967
2012 January	0.6658
2012 February	0.6258
2012 March	0.6102
2012 April	0.6364
2012 May	0.682
2012 June	0.7049
2012 July	0.7007
2012 August	0.7055
2012 September	0.6967
2012 October	0.6716

Year Months	Value
2012 November	0.6764
2012 December	0.6517

